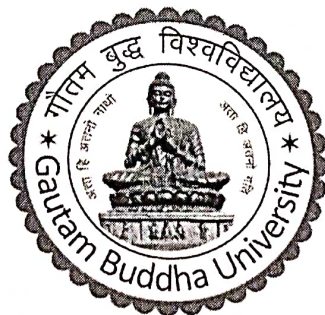


**UNIVERSITY SCHOOL
OF
INFORMATION AND COMMUNICATION TECHNOLOGY**
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND 4 IR TECHNOLOGIES

PROGRAMME STRUCTURE

M.TECH. CYBER SECURITY

2021-2023



GAUTAM BUDDHA UNIVERSITY
GAUTAM BUDH NAGAR, GREATER NOIDA, UP, INDIA

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SEMESTER I

S.No.	Course Code	Course Name	L	T	P	Credits	Types
1	CS521	Advanced Data Base Management System	3	0	0	3	CC1
2	CS523	Design and Architecture for Software Systems	3	0	0	3	CC2 / FC
3	CS525	Advanced Data Structure and Algorithm	3	1	0	4	CC3
4	CS527	Research Techniques in ICT	3	0	0	3	CC4
5	CS529	Java Programming	3	0	0	3	CC5
6	ES415	Energy and Environment	3	0	0	3	OE1 / AECC
7	EN531	Language Culture and Society	3	0	0	3	OE2 / AECC
8	CS581	Advanced Data Base Management System Lab	0	0	3	2	CC-L1
9	CS583	Java Programming Lab	0	0	3	2	CC-L2 / SEC
10	GP	General Proficiency	Non Credit				
Total Hours and Credits			21	1	6	26	

SEMESTER II

S.No.	Course Code	Course Name	L	T	P	Credits	Types
1	CS522	Python Programming	3	0	0	3	CC6
2	CC522	Fundamental of Cyber Security	3	0	0	3	CC7
3	CC524	Advanced Principles of Secure Coding	3	0	0	3	CC8
4	CC526	Digital Forensics Techniques	3	0	0	3	CC9
5		Elective-1	3	0	0	3	E1 / DSE
6		Elective-2	3	0	0	3	E2 / DSE
7		Generic Elective	3	1	0	4	GE1
8	CS582	Python Programming Lab	0	0	3	2	CC-L3 / SEC
9	CC582	Cyber Security Lab	0	0	3	2	CC-L4
10	GP	General Proficiency	Non Credit				
Total Hours and Credits			21	1	6	26	

Summer Project in Industry will be done individually after Fourth Year of Integrated B.Tech. - M.Tech. CSE and Second Year of M.Tech. during the summer break and it will be of minimum 4 weeks. It will be evaluated as per University Examination Rules in III semester of M.Tech. and IX semester for Integrated B.Tech. - M.Tech. CSE.

SEMESTER III

S.No.	Course Code	Course Name	L	T	P	Credits	Types
1	CC621	Digital Forensics and Incident Response	3	0	0	3	CC10
2	CC623	Ethical Hacking and Penetration Testing	3	0	0	3	CC11
3		Elective-3	3	0	0	3	E3 / DSE
4		Elective-4	3	0	0	3	E4 / DSE
5	CC681	Digital Forensics Lab	0	0	3	2	CC-L5
6	CC683	Summer Project	0	0	8	4	SP / E
7	CC691	Dissertation Part - I	0	0	16	8	DP1 / E
8	GP	General Proficiency	Non Credit				
Total Hours and Credits			12	0	27	26	

SEMESTER IV

S.No.	Course Code	Course Name	L	T	P	Credits	Types
1	CC692	Dissertation Part - II	0	0	52	26	DP2 / E
2	GP	General Proficiency	Non Credit				
Total Hours and Credits			0	0	52	26	

GRAND TOTAL OF CREDITS = 104

Summer Project will be done individually and It will be evaluated as per University Examination Rules.

Dissertation will be done individually and It will be evaluated as per University Examination Rules.

USICT will provide a mentor/supervisor for summer project, and dissertation.

ELECTIVES FROM AI4IRT

S.No.	Course Code	Course Name	L	T	P	Credits	Types
1	CC528	Operating System Security	3	0	0	3	E1
2	CC530	Image and Video Processing Techniques	3	0	0	3	E1
3	CC532	Access Control and Identity Management Systems	3	0	0	3	E1
4	CC534	Physical Security of IT Infrastructure	3	0	0	3	E1
5	CC536	Applied Cryptography	3	0	0	3	E2
6	CC538	Parallel and Distributed Computing	3	0	0	3	E2
7	CC540	Cloud Architectures and Security	3	0	0	3	E2
8	CC542	Advanced Security with Web Technology	3	0	0	3	E2
9	CC635	Data Security and Privacy	3	0	0	3	E3
10	CC637	Android Security Design and Internals	3	0	0	3	E3
11	CC639	Biometric Systems and Security	3	0	0	3	E3
12	CC631	Blockchain Technology	3	0	0	3	E3
13	CC633	Information Warfare	3	0	0	3	E4
14	CC635	Risk Management and Audit	3	0	0	3	E4
15	CC637	Mobile and Wireless Network Security	3	0	0	3	E4
16	CC639	Machine Learning Techniques	3	0	0	3	E4

OPEN AND GENERIC ELECTIVES FROM OTHER SCHOOLS

17	ES415	Energy and Environment	3	0	0	3	OE1
18	EN531	Language, Culture and Society	3	0	0	3	OE2
19	MA402	Modeling and Simulation	3	1	0	4	GE1
20	MA416	Probability and Stochastic Process	3	1	0	4	GE1

CC Computer Science and Cyber Security for Course Code

CS Computer Science for Course Code

CC Core Course from USICT for Type of Course

CC-L Core Course Lab from USICT for Type of Course

GE General Elective from related discipline of other Deptt./School

AECC Ability Enhancement Compulsary Course

OE Open Elective from other discipline of other Deptt./School

FC Foundation Course

DSC Discipline Specific Course

SEC Skill Enhancement Course

E Elective

DP1 Dissertation Part 1

DP2 Dissertation Part 2

SP Summer Project